7.4 Hydroxy Compounds (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	7. Organic Chemistry (A Level Only)
Topic	7.4 Hydroxy Compounds (A Level Only)
Difficulty	Medium

Time allowed: 70

Score: /51

Percentage: /100

Question 1a

This question is about the reactions of the aromatic alcohol, phenol.

Phenol behaves as a weak acid.

i)
Write the equation for phenol reacting with sodium hydroxide.

[2]

ii)
Write the equation for phenol reacting with sodium metal.

[3]

[5 marks]

Question 1b

Phenols also undergo electrophilic substitution reactions when reacted with bromine water at room temperature.

Describe two observations that can be made during this reaction.

[1 mark]

Question 1c

Draw the structures of the two possible products from the reaction of phenol with **dilute** nitric acid.

[2 marks]

Question 1d

Explain why 3–nitrophenol is not formed when phenol reacts with dilute nitric acid.

[1 mark]

Question 2a

This question is about phenol.

Phenol is an aryl compound.

i)
Give the molecular formula of phenol.

[1]

ii)
Give the empirical formula of phenol.

[1]

[2 marks]

Question 2b

Explain why phenol is a weak acid.

You should include an equation in your answer.

[3 marks]

Question 2c

Describe and explain how the acidities of ethanol and phenol compare to that of water.

[4 marks]

Question 2d

Phenol reacts with bromine water.

i)
Give the name of the product formed by this reaction.

[1]

ii)
Write a balanced chemical equation for this reaction.

[1]

[1 mark]

Question 2e

Explain how the reactions of phenol and benzene with bromine can show that phenol is more reactive than benzene.

[2 marks]

Question 3a

Phenol can be prepared from phenylamine

There are three stages that take place during this preparation:

- **Stage 1:** Prepare nitric(III) acid using sodium nitrate and hydrochloric acid
- **Stage 2:** React phenylamine with nitric(III) acid and hydrochloric acid to form a diazonium salt
- **Stage 3:** Thermal decomposition of the diazonium salt to form phenol, hydrochloric acid, and nitrogen gas

i)
Give the essential conditions for the reaction in **stage 2.**

[1]

ii)
Draw the displayed formula of the diazonium salt formed in **stage 2.**

[3]

[4 marks]

Question 3b

Write balanced symbol equations for the reactions that occur at each stage:

Stage 1: ....
Stage 2: ....
Stage 3: ....

[3 marks]

Question 3c

The diazonium salt formed during stage 2 can also react with phenol to form a useful substance, **Z**.

i)
Give the displayed formula of **Z.**

[2]

ii)
Suggest a possible use for **Z**.

[1]

[3 marks]

Question 3d

The first step of substance Z involves creating the diazonium ion.

The reaction is the same as that in stage 2 in part **(a)**.

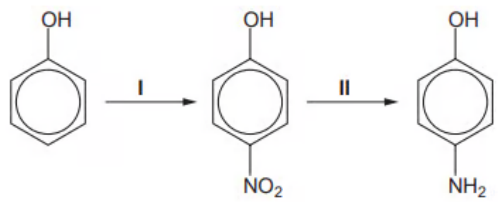
Explain why the reaction to produce the diazonium ion is carried out at below 10 °C.

[1 mark]

Question 4a

Rodinol is used as a photographic developer.

It can be synthesised from phenol by the following route:



i)
Suggest reagents and conditions for step 1.

[1]

ii)
Name the mechanism by which this reaction occurs.

[1]

[2 marks]

Question 4b

i)
What type of reaction is step II?

[1]

ii)
Give suitable reagents that could be used in step 2.

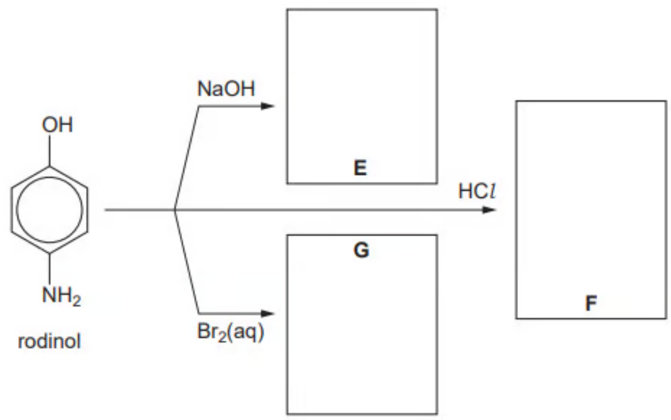
[2]

[3 marks]

Question 4c

Rodinol can undergo many reactions.

Suggest the structural formulae for the compounds **E**, **F** and **G** in the following chart of the reaction so of rodinol.



[3 marks]

Question 5a

Acyl chlorides are useful reagents in synthesis.

They react with aromatic compounds and also with alcohols.

i)
Give the reagents and conditions required to make ethyl ethanoate directly from a carboxylic acid.

[3]

ii)
Write an equation to show the formation of ethyl ethanoate in part i).

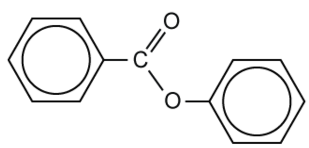
[1]

[4 marks]

Question 5b

Phenyl benzoate is made from the reaction between benzoyl chloride and phenol.

Its structure is shown below.



What is the molecular formula of phenyl benzoate?

[1 mark]

Question 5c

i)
Give the essential conditions needed for the reaction in part **(b)**.

[2]

ii)
Describe one observation you would make in this reaction.

[1]

[3 marks]

Question 5d

During the initial reaction to produce phenyl benzoate, the phenoxide ion is formed.

i)

Draw the structure of this ion.

[1]

ii)

Explain why it this ion is formed.

[1]

iii)

Write an equation for the overall reaction.

[1]

[3 marks]

Model Answers: Medium

1a

a)

i) The equation for phenol reacting with sodium hydroxide is:

$$C_6H_5OH + NaOH \rightarrow C_6H_5O^-Na^+ + H_2O$$

- Correct reactants; [1 mark]
- Correct products; [1 mark]

ii) The equation for phenol reacting with sodium metal is:

$$2C_6H_5OH + 2Na \rightarrow 2C_6H_5O^-Na^+ + H_2;$$

- Correct reactants; [1 mark]
- Correct products; [1 mark]
- Correctly balanced; [1 mark]

[Total: 5 marks]

- These reactions are neutralisation reactions
    - Acid + Base → Salt + Water
    - Acid + Metal → Salt + Hydrogen gas
- The product should be $C_6H_5O^-Na^+$ as the phenoxide ion is $C_6H_5O^-$ and the sodium ion is Na^+ meaning there is more of an ionic character than covalent character to the bond

1b

b) Two observations for the reaction between bromine and phenol are:

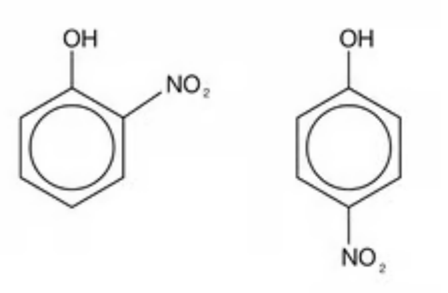
- Bromine water decolourises; [1 mark]
- A white precipitate formed; [1 mark]

[Total: 2 marks]

- Phenol decolourises the orange bromine solution to form a white precipitate of 2,4,6–tribromophenol

1c

c) The structures of the two possible products from the reaction of phenol with **dilute** nitric acid are:



- Each correct structure scores [1 mark]

[Total: 2 marks]

- You should know the products when phenol reacts separately with bromine and **dilute** nitric acid
- When concentrated HNO_3 is used, the product will be 2,4,6–trinitrophenol instead

1d

d) 3–nitrophenol is not formed because:

- The- OH group is 2,4, 6directing; [1 mark]

[Total: 1 mark]

- The O atom from the OH group donates electron density into the electron ring making it more susceptible from attack by electrophiles
- The incoming electrophiles are directly by the OH group to the 2,4, and 6 positions

2a

a)

i)The molecular formula of phenol is:

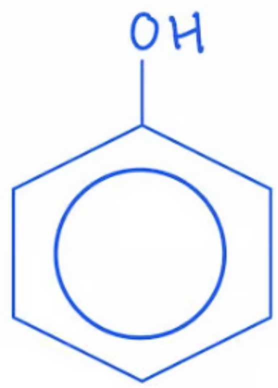
- C_6H_6O; [1 mark]

ii) The empirical formula of phenol is:

- C_6H_6O; [1 mark]

[Total: 2 marks]

- The structure of phenol is:



- Sometimes it is easier to draw it out with all atoms showing so that you don't miscount the number of hydrogen atoms
- The molecular formula is the actual number of atoms of each element in a compound
- The empirical formula is the simplest whole number ratio of atoms of each element
- The molecular formula cannot be simplified so is the same as the empirical formula

2b

b) Phenol can acts as a weak acid because:

- Phenol can donate the H in the OH group; [1 mark]
- $C_6H_5OH + H_2O \leftrightharpoons C_6H_5O^- + H_3O^+$
  OR
  $C_6H_5OH \leftrightharpoons C_6H_5O^- + H^+$; [1 mark]

- Phenoxide ion is stable because its negative charge is spread out (over the entire ion); [1 mark]

[Total: 3 marks]

- Acids donate protons
- Although a weak acid, phenol can still lose its H atom from the -OH group
- The phenoxide ion is more stable than phenol so the position of the equilibrium in the equation below lies to the right hand side making it more likely to act as an acid than a base
  - $C_6H_5OH \leftrightharpoons C_6H_5O^- + H^+$

2c

c) Describe and explain how the acidities of ethanol and phenol compare to that of water:

- Ethanol is less acidic than water; [1 mark]
- (Because) the ethyl group concentrates more negative charge on the oxygen atom (of the OH) group; [1 mark]
- Phenol is more acidic than water; [1 mark]
- (Because) the anion formed is more stable (due to the spread of charge); [1 mark]

[Total: 4 marks]

- The question states to compare the acidities to water so you must be explicit about this
- The order of acidity for all three, beginning with the least acidic is:
  - ethanol < water < phenol
- The level of acidity is related to how well the anion formed keeps hold of the H atom from the OH group
  - In phenol, the ion formed is $C_6H_5O^-$
  - In water the ion formed is OH^-
  - In ethanol, the ion formed is $C_2H_5O^-$
- The phenoxide ion is the only ion with its charge spread across the whole ion
  - The charge density of the negative charge on the phenoxide ion is reduced so H^+ ions are not strongly attracted to the phenoxide ion meaning it is less likely to reform the phenol molecule
  - In ethanol, the ethyl group is electron donating so concentrates more

negative charge on the oxygen atom in the $C_6H_5O^-$ ion meaning it more

- (Because) the anion formed is more stable (due to the spread of charge); [1 mark]

[Total: 4 marks]

- The question states to compare the acidities to water so you must be explicit about this
- The order of acidity for all three, beginning with the least acidic is:
  - ethanol < water < phenol
- The level of acidity is related to how well the anion formed keeps hold of the H atom from the OH group
  - In phenol, the ion formed is $C_6H_5O^-$
  - In water the ion formed is OH^-
  - In ethanol, the ion formed is $C_2H_5O^-$
- The phenoxide ion is the only ion with its charge spread across the whole ion
  - The charge density of the negative charge on the phenoxide ion is reduced so H^+ ions are not strongly attracted to the phenoxide ion meaning it is less likely to reform the phenol molecule
  - In ethanol, the ethyl group is electron donating so concentrates more negative charge on the oxygen atom in the $C_2H_5O^-$ ion meaning it more readily accepts H^+

2d

d)

i) The name of the product formed by this reaction is:

- 2,4,6–tribromophenol; [1 mark]

ii) A balanced chemical equation for this reaction is:

- $3Br_2 + C_6H_5OH \rightarrow C_6H_2Br_3OH + 3HBr$; [1 mark]

[Total: 2 marks]

- **Remember:** The -OH group 'activates' the benzene ring and directs substitution of groups onto positions 2,4, and 6
- The product is observed as a white precipitate

2e

e) This reaction can be used to show that phenol is more reactive than bromine because:

Any **two** from:

- Benzene only reacts with liquid / pure bromine but not with bromine water; [1 mark]
- Benzene monosubstitutes while phenol trisubstitutes; [1 mark]
- Benzene needs catalyst / $AlBr_3$ catalyst; [1 mark]

[Total: 3 marks]

- You must be able to compare the bromination and nitration of phenol and benzene
- The reaction of benzene with bromine will only take place at room temperature in the presence of a catalyst, whereas phenol does not require the catalyst

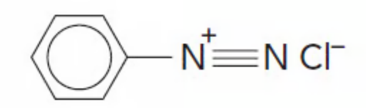
3a

a)

i) The essential conditions for the reaction in **stage 2** are:

- Temperature below 10 °C; [1 mark]

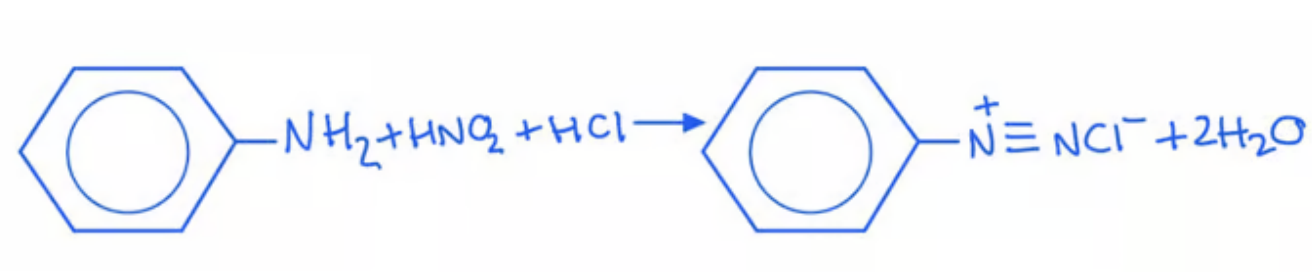
ii) The displayed formula for the diazonium salt formed in **stage 2** is:



- C_6H_5- group; [1 mark]
- N ≡ N; [1 mark]
- + charge and Cl^- ion; [1 mark]

[Total: 4 marks]

- The temperature is kept below 10 °C to prevent the diazonium salt decomposing prematurely
- The diazonium salt formed in step 2 is benzenediazonium chloride
- The following reaction happens:



3b

b) The balanced symbol equations are:

- Stage 1: $NaNO_2$ + HCl → HNO_2 + NaCl; [1 mark]
- Stage 2: $C_6H_5NH_2$ + HNO_2 + HCl → $C_6H_5N_2^+Cl^-$ + H_2O
  OR
  $C_6H_5NH_2$ + $NaNO_2$ + 2 HCl → $C_6H_5N_2^+Cl^-$ + $2H_2O$; [1 mark]
- Stage 3: $C_6H_5N_2^+Cl^-$ + H_2O → C_6H_5OH + HCl + N_2 ; [1 mark]

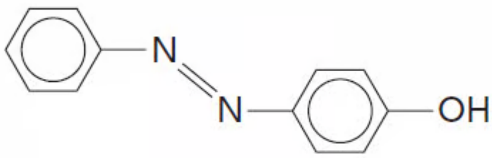
[Total: 3 marks]

- You need to be able to recall the steps and equations involved in the preparation of phenol but in this case you have been given some information
- **Stage 1**:
  - You are given both reactants and one product
  - Sodium nitrate + hydrochloric acid → nitric(III) acid
  - It is logical to deduce that the sodium and chlorine will react to produce sodium chloride as the second product
  - Sodium nitrate + hydrochloric acid → nitric(III) acid + sodium chloride
- **Stage 2**:
  - Again, you are given all three reactants
  - A salt is formed in a neutralisation reaction so water is the second product
  - Phenylamine + nitric(III) acid + hydrochloric acid → benzenediazonium chloride + water
- **Stage 3:**
  - Benzenediazonium chloride decomposes when warmed with water
  - Benzenediazonium chloride + water → phenol + hydrochloric acid + nitrogen

3c

c)

i) The displayed formula of **Z** is:



- Arene rings and phenol group; [1 mark]
- Presence of -N = N-; [1 mark]

ii) A possible use of **Z** is:

- Dye; [1 mark]

[Total: 3 marks]

- Azo dyes are formed from the reaction between diazonium ions and phenol in a coupling reaction
- In this particular reaction, an orange dye will form
- Make sure you are able to draw this particular azo dye

3d

d) The reaction to produce the diazonium ion is carried out at below 10 ^{o}C because:

- The diaznonium ion will decompose / to prevent the diazonium ion decomposing; [1 mark]

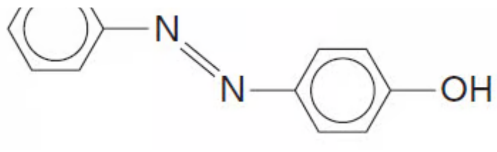
[Total: 1 mark]

- In Step 2 of creating substance Z, the diazonium ion is reacted with phenol
- The lower temperatures prevent the ion from decomposing before this reaction can happen
- The decomposition of the diaznonium ion would produce nitrogen gas

4a

a)

i) Reagents and conditions for step 1 is:



- Arene rings and phenol group; [1 mark]
- Presence of -N = N-; [1 mark]

ii) A possible use of **Z** is:

- Dye; [1 mark]

[Total: 3 marks]

- Azo dyes are formed from the reaction between diazonium ions and phenol in a coupling reaction
- In this particular reaction, an orange dye will form
- Make sure you are able to draw this particular azo dye

3d

d) The reaction to produce the diazonium ion is carried out at below 10 °C because:

- The diaznonium ion will decompose / to prevent the diazonium ion decomposing; [1 mark]

[Total: 1 mark]

- In Step 2 of creating substance Z, the diazonium ion is reacted with phenol
- The lower temperatures prevent the ion from decomposing before this reaction can happen
- The decomposition of the diaznonium ion would produce nitrogen gas

4a

a)

i) Reagents and conditions for step 1 is:

- Dilute HNO_3 / HNO_3 (aq); [1 mark]

ii)The mechanism by which this reaction occurs is:

- Electrophilic substitution; [1 mark]

[Total: 2 marks]

- You will not score the mark for writing concentrated HNO_3
- The product shown in the sequence has the nitro group at position 4
- This is produced when dilute nitric acid is used
- If you had used concentrated nitric acid, there would be nitro groups at positions 2,4, and 6 and the compound 2,4,6–trinitrophenol produced

4b

b)

i) The type of reaction in step II is:

- Reduction; [1 mark]

ii) The suitable reagents that could be used in step II are:

- Sn / conc HCl; [1 mark]
- Reflux; [1 mark]

[Total: 3 marks]

- The 4–nitrophenol gains electrons from the tin in the presence of the conc HCl
- It is important that you write both conditions for the two marks

4c

c) The structural formulae for **E**, **F,** and **G** are:

- **E** = $H_2N\text{-}C_6H_4\text{-}O^- Na^+$ / $H_2N\text{-}C_6H_4\text{-}ONa$; [1 mark]
- **F**= $HO\text{-}C_6H_4NH_3{}^+Cl^-$ / $HO\text{-}C_6H_4NH_3Cl$; [1 mark]
- **G** = $HO\text{-}C_6H_2Br_2\text{-}NH_2$ / $HO\text{-}C_6HBr_3\text{-}NH_2$ / $HO\text{-}C_6Br_4\text{-}NH_2$; [1 mark]

[Total: 3 marks]

- Rodinol is a weak acid so can react with bases such as sodium hydroxide.
  - As well as $H_2N\text{-}C_6H_4\text{-}ONa$, water is also produced
- The reaction with aqueous bromine will result in bromine substituting the

hydrogen atoms
- The most likely product when reacting bromine with rodinol is HO-$C_6H_2Br_2$-NH_2 because the bromine is directed to the 2 and 6 positions but it is possible to have bromines subsituted at other positions to give HO-C_6HBr_3-NH_2 or HO-C_6Br_4-NH_2

5a

a)

i) The reagents and conditions required to make ethyl ethanoate directly from a carboxylic acid are:

- Ethanol; [1 mark]
- Ethanoic acid; [1 mark]
- conc H_2SO_4; [1 mark]

ii) An equation to show the formation of ethyl ethanoate in part i) is:

- $C_2H_5OH + CH_3COOH \rightleftharpoons CH_3COOC_2H_5 + H_2O$; [1 mark

[Total: 4 marks]

- This should be a straightforward question
- An alcohol and carboxylic acid form an ester and water
- Conc sulfuric acid is used as a catalyst

5b

b) The molecular formula of phenyl benzoate is:

- $C_{13}H_{10}O_2$; [1 mark]

[Total: 1 mark]

- To determine the molecular formula, count the number of atoms of each element
- Some students find it useful to draw out the carbon and hydrogen atoms fully so that they don't miscount them

5c

b)

i) The reagents and conditions needed for the reaction in part **(b)** are:

- Warm; [1 mark]
- Base / NaOH; [1 mark]

ii) One observations you would make in this reaction is:

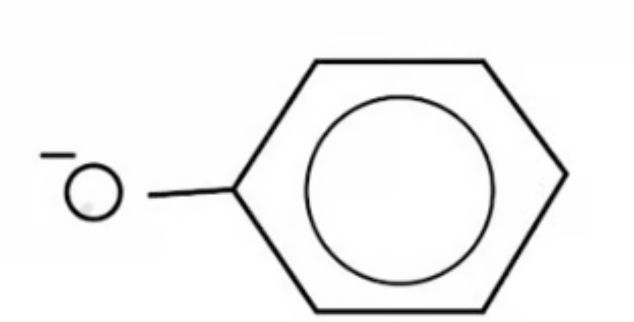
- (Steamy) white fumes (of hydrogen chloride); [1 mark]

[Total: 3 marks]

5d

d)

i) The structure of the phenoxide ion is:



- Correct structure; [1 mark]

ii) This ion is formed because:

- It acts as a better nucleophile (than phenol); [1 mark]

iii) An equation for the overall reaction is:

- $C_6H_5OH + C_6H_5COCl \rightarrow C_6H_5COOC_6H_5 + HCl$; [1 mark]

[Total: 3 marks]

ii) The observations you would make in this reaction is:

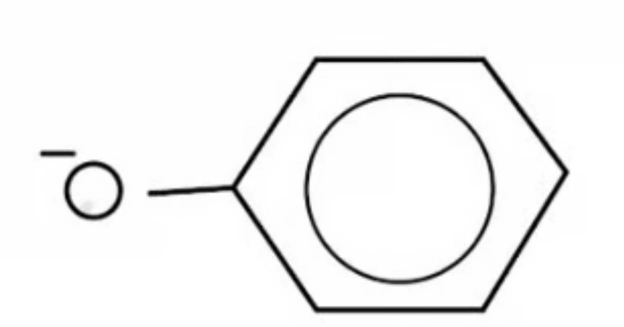
- (Steamy) white fumes (of hydrogen chloride); [1 mark]

[Total: 3 marks]

5d

d)

i) The structure of the phenoxide ion is:



- Correct structure; [1 mark]

ii) This ion is formed because:

- It acts as a better nucleophile (than phenol); [1 mark]

iii) An equation for the overall reaction is:

- $C_6H_5OH + C_6H_5COCl \rightarrow C_6H_5COOC_6H_5 + HCl$; [1 mark]

[Total: 3 marks]

- The initial reacction between the phenol and base forms the phenoxide ion, $C_6H_5O^-$
- Being a better nucleophile means it is better at attacking the carbonyl carbon in the acyl chloride
- You must have the negative charge on the oxygen atom to score the mark but but its location doesn't matter